

PROJECT PROPOSAL
GROUP PROJECT
CST 394-2

Group No: CST 09

TileVista:

Virtual Tile and Bathware Visualization &
Showroom Management System

Computer Science and Technology Degree Program

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University

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Declaration

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Dr. H.M.S.N. Ariyadasa, Department of Computer Science and Informatics, Uva Wellassa University of Sri Lanka and Mr. K.R.T.A. Ekanayaka, Department of Computer Science and Informatics, Uva Wellassa University of Sri Lanka. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit CST 394-2, Project II for the degree of Computer Science and Technology Degree Program.

Group Details:

Group No: CST 09

No.	Name of the Student	Index Number	E-mail address	Signature
01	R.H.I.S Kularathne	UWU/CST/22/006	cst22006@std.uwu.ac.lk	
02	D.P.K.Shehara	UWU/CST/22/030	cst22030@std.uwu.ac.lk	
03	A.N.S.L.Alahapperuma	UWU/CST/22/035	cst22035@std.uwu.ac.lk	
04	R.G.S.N Gunathilake	UWU/CST/22/057	cst22057@std.uwu.ac.lk	

Supervisor Details:

Name of the Supervisor	E-mail address	Contact Number
Dr.H.M.S.N. Ariyadasa	subhash@uwu.ac.lk	+94 55 35 600 90

Name of the Co-supervisor	E-mail address	Contact Number
Mr. K.R.T.A. Ekanayaka	1257tharinduashan@gmail.com	+94 55 35 600 90

Project Approval:

.....
Dr.H.M.S.N. Ariyadasa

Project Supervisor

Date:

.....

.....
Mr. K.R.T.A. Ekanayaka

Project Co-supervisor

Date:

.....

Table of Contents

1	Introduction	1
1.1	Project Title	1
1.2	Project Description.....	1
1.3	Background and Motivation.....	2
1.3.1	Background.....	2
1.3.2	Motivation.....	3
1.4	Problem in Brief	4
1.4.1	Customer-Side Problems	4
1.4.2	Business-Side Problems.....	5
1.5	Proposed Solution	7
1.6	Project Aim and Objectives.....	9
1.6.1	Aim	9
1.6.2	Objectives	9
2	Methodology.....	10
2.1	Agile Methodology – Scrum Framework.....	10
2.1.1	Sprint 1:.....	11
2.1.2	Sprint 2:.....	13
2.1.3	Sprint 3:.....	15
2.1.4	Sprint 4:.....	17
3	Requirements Identification	19
3.1	Functional and Non-functional requirements.....	19
3.1.1	Functional requirements.....	19
3.1.2	Non-functional requirements	22
3.2	System requirements	24
3.2.1	Hardware Requirements.....	24
3.2.2	Software Requirements	24
3.2.3	Technologies	25
3.3	User roles.....	26
4	Project Plan.....	28
4.1	Gantt chart	28
5	References	29

List of Figures

Figure 1 - The Agile–Scrum Framework.....	10
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List of Tables

Table 1 - Gantt Chart	28
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1 Introduction

1.1 Project Title

TileVista

1.2 Project Description

The proposed system is a web-based platform developed for “Alahapperuma Trade Center,” a tile and bathware showroom, with the aim of enhancing the customer purchasing experience and improving business decision-making. The system provides an interactive and user-friendly environment where customers can explore, select, and visualize products before making a purchase.

The platform offers a wide range of product categories including tiles, bathware, accessories, and pre-designed packages. Customers can browse these products using advanced filtering options such as type, color, material, brand, size, and price. Each product is displayed with detailed information including price, available quantity, brand name, and any applicable discounts, allowing users to make informed decisions.

A key feature of the system is the Customer-Based 3D Bathroom Designer, which enables users to create and customize a 3D bathroom environment according to their own bathroom layout and preferences. Customers can select the bathroom shape and size, enter room dimensions, add doors and windows, customize wall colors, apply floor and wall tile designs, and place bathware items such as bathtubs, wash basins, toilets, showers, mirrors, and accessories into the space. Users can also move, rotate, replace, and remove selected items while using zoom and rotation controls to inspect the design from different angles. This feature helps customers create a more personalized bathroom design, reduce uncertainty, and choose suitable products with greater confidence.

In addition to individual product selection, the system also promotes pre-designed packages that include a combination of items at discounted prices. These packages can be visualized within the 3D bathroom designer, making it easier for customers to explore complete solutions based on their budget and preferences.

From the business perspective, the system integrates with the showroom's existing Point of Sale (POS) system to provide real-time inventory updates. An admin dashboard is included to analyze product performance, track sales trends, identify fast-moving and slow-moving items, and support decision-making related to restocking and promotional strategies. This reduces the need for manual analysis and improves operational efficiency.

The system is primarily designed for customers who are planning to purchase tiles and bathroom products, as well as for shop owners and administrators who manage inventory and sales. By combining product visualization, smart selection tools, and data-driven insights, the system aims to improve customer satisfaction, reduce incorrect purchases, and enhance overall business performance.

1.3 Background and Motivation

1.3.1 Background

In recent years, the retail industry, especially in construction and home improvement sectors such as tiles and bathware, has been gradually shifting towards digital platforms. Customers increasingly prefer to explore products online before visiting physical stores, as it saves time and provides access to a wider range of options. However, unlike other e-commerce domains, tile and bathware selection involves a strong visual and spatial component, making it difficult for customers to make confident decisions based solely on images and descriptions.

Traditional showroom experiences attempt to address this issue by displaying sample setups, but these are limited in variety and cannot represent every possible combination of products. As a result, customers often rely on imagination when selecting tiles and bathroom items, which can lead to dissatisfaction after installation. Studies have shown that lack of visualization tools in online retail environments reduces customer confidence and increases the likelihood of product returns [1].

With the advancement of modern web technologies such as 3D visualization and interactive rendering, many modern systems now incorporate virtual design tools to enhance user experience. Technologies like Three.js and WebGL-based frameworks enable interactive product visualization directly within web browsers, allowing

customers to explore and customize virtual environments in real time. Additionally, data-driven decision-making has become essential for businesses, where analytics dashboards are used to monitor sales trends, manage inventory, and optimize pricing strategies [2].

These developments highlight the growing need for an integrated system that combines product visualization, smart selection tools, and business analytics within a single platform, particularly for industries like tile and bathware retail.

1.3.2 Motivation

The motivation for this project arises from the practical challenges observed in tile and bathware showrooms, particularly in local businesses such as “Alahapperuma Trade Center.” Customers often face difficulty in selecting suitable products due to the inability to visualize how different tiles and bathroom items will appear within a bathroom layout similar to their own space. This uncertainty leads to confusion, time-consuming decision-making, and sometimes incorrect purchases, resulting in dissatisfaction and potential returns.

At the same time, shop owners rely heavily on manual methods to track inventory, analyze sales, and decide on discounts or restocking. This approach is not only time-consuming but also lacks accuracy, making it difficult to identify fast-moving products, slow-selling items, and effective marketing strategies.

Inspired by modern e-commerce platforms and emerging technologies in 3D visualization and smart systems, this project aims to bridge the gap between customer expectations and existing retail practices. The idea of integrating a Customer-Based 3D Bathroom Designer was influenced by current trends in immersive shopping experiences, where users can interact with products in a simulated environment before purchasing [3]. This feature allows customers to customize bathroom dimensions, layout elements, tiles, and showroom products in a more personalized 3D environment. Furthermore, incorporating an analytics dashboard is motivated by the increasing importance of data-driven decision-making in improving business efficiency and competitiveness [2].

By combining these concepts, the project seeks to create a system that not only enhances the customer experience but also supports business growth. The ultimate motivation is to provide a practical, technology-driven solution that reduces uncertainty in purchasing, improves decision-making, and brings innovation to traditional showroom operations.

1.4 Problem in Brief

The current tile and bathware purchasing process, especially in traditional showroom environments, presents several challenges for both customers and business owners. These issues affect decision-making, customer satisfaction, and overall business efficiency. The key problems identified are as follows:

1.4.1 Customer-Side Problems

1. Lack of Visualization

Customers are unable to clearly see how selected tiles and bathware items will look when combined within a bathroom layout similar to their own space. Since most showrooms only display limited sample setups, customers cannot visualize different combinations of colors, textures, and layouts. This limitation reduces confidence in decision-making and affects the overall purchasing experience [3].

2. Dependence on Imagination

In the absence of interactive tools, customers must rely on their imagination to predict the final appearance of their space. However, human imagination is often inaccurate when dealing with spatial design and color combinations, leading to uncertainty and hesitation during product selection.

3. Difficulty in Matching Products

Selecting tiles, wall colors, and bathroom accessories that match well together is a complex task. Customers may lack design knowledge, making it difficult to achieve a visually appealing combination. This often results in mismatched designs and unsatisfactory outcomes.

4. Confusion Due to Wide Variety

Modern showrooms offer a large number of products with different styles, sizes, colors, and brands. While variety is beneficial, it can overwhelm customers, making it harder to compare options and choose the most suitable product [4].

5. Time-Consuming Selection Process

Customers often spend a significant amount of time visiting showrooms, comparing samples, and discussing options with staff. This process becomes inefficient, especially when customers are unsure about their choices.

6. Incorrect Purchases and Dissatisfaction

Due to the lack of proper visualization and the heavy reliance on imagination, customers may select tiles or bathware items that do not match well with their expected design after installation. This creates several difficulties for customers:

- Customers may feel disappointed when the final appearance does not match their expectations.
- Returning or replacing products wastes customer time and effort, as they need to revisit the showroom and repeat the selection process.
- Additional expenses may occur due to transportation costs, labor charges, or price differences between replacement items.
- The overall purchasing experience becomes stressful and inconvenient, reducing customer satisfaction and confidence.

1.4.2 Business-Side Problems

1. Lack of Interactive Customer Experience

Traditional showrooms rely mainly on physical displays, which provide limited interaction. Without digital tools such as virtual visualization, businesses fail to engage customers effectively or provide a modern shopping experience [5].

2. Manual Inventory Management

Many small and medium-scale businesses still manage inventory manually or with basic systems. This increases the risk of errors, outdated information, and inefficient tracking of product availability.

3. Difficulty in Identifying Product Performance

Shop owners often do not have clear insights into which products are selling quickly and which are not. Without proper analytics, it becomes difficult to plan inventory and marketing strategies effectively.

4. Ineffective Decision Making

Decisions related to restocking, pricing, and discounts are often based on experience rather than data. This can lead to overstocking of low-demand items or shortages of high-demand products, affecting profitability [6].

5. Underutilization of Product Packages

Although showrooms offer bundled packages with discounts, customers may not fully understand their value due to lack of proper presentation and visualization. This results in missed opportunities to increase sales.

6. Limited Use of Modern Technologies

Many traditional businesses have not yet adopted advanced technologies such as 3D visualization, virtual design tools, and analytics dashboards. This puts them at a disadvantage compared to modern digital competitors.

7. Incorrect Purchases and Product Returns

Incorrect purchases and product returns also create challenges for showroom owners and management:

- Handling returned or replaced items increases the workload of showroom staff.
- Returned products may become difficult to resell, especially if packaging is damaged or items are customized.
- Frequent returns can lead to financial losses and operational inefficiencies.
- Customer dissatisfaction may negatively affect the reputation and trustworthiness of the business.

- Additional time and resources are required to manage replacements, inventory updates, and customer complaints.

1.5 Proposed Solution

To overcome the identified problems in traditional tile and bathware showrooms, the proposed system introduces a modern web-based platform for “Alahapperuma Trade Center.” The system combines interactive product visualization, smart product selection, customized 3D bathroom designing, and business analytics to improve both customer experience and showroom management. The proposed solution focuses on reducing customer uncertainty during purchasing while supporting efficient and data-driven business operations.

The main proposed solutions of the system are as follows:

1. Interactive Product Browsing
 - Provide an organized platform where customers can browse tiles, bathware items, accessories, and packages with detailed product information.
2. Advanced Product Filtering
 - Allow customers to filter products based on category, color, material, brand, size, and price to simplify the selection process.
3. Customer-Based 3D Bathroom Designer
 - Provide a customizable 3D bathroom design environment where customers can create a bathroom layout based on their own preferences by selecting bathroom shape, size, dimensions, doors, windows, wall colors, floor tiles, wall tiles, and showroom products.
4. Product Placement and Room Customization
 - Allow customers to add bathware items and accessories such as bathtubs, wash basins, toilets, showers, mirrors, and other available showroom items into the 3D bathroom. Users can move, rotate, replace, and remove selected items according to their layout.

5. Interactive 3D View and Measurement Support
 - Enable users to zoom, rotate, and inspect the customized 3D bathroom from different angles. A ruler or measurement tool will also be provided to help users check space availability and item placement distances.
6. Package Visualization
 - Display showroom packages within the customized 3D bathroom environment so customers can easily explore complete product combinations and discounted offers.
7. Price Summary for Selected Items
 - Automatically calculate and display the total price summary for selected tiles, bathware items, accessories, and packages to support customer purchasing decisions.
8. Real-Time Inventory Integration
 - Connect the system with the showroom's POS database to display accurate product availability and quantity updates.
9. Analytics Dashboard for Management
 - Provide sales and inventory analysis to help administrators identify fast-selling products, slow-moving items, sales trends, and restocking requirements.
10. Decision Support for Business
 - Growth Support showroom management in planning discounts, promotions, and inventory strategies through data-driven insights.
11. Reduction of Incorrect Purchases and Returns
 - Improve customer confidence through customized 3D visualization features, helping users make better purchasing decisions and reducing dissatisfaction after installation.

1.6 Project Aim and Objectives

1.6.1 Aim

To provide an interactive web-based platform that enhances the customer purchasing experience through a customer-based 3D bathroom designing feature, simplifies product selection with smart filtering options, and supports efficient showroom management using real-time inventory integration and analytical insights.

1.6.2 Objectives

1. To develop a user-friendly web platform that allows customers to browse, filter, and explore tiles, bathware items, accessories, and showroom packages based on their preferences.
2. To implement a Customer-Based 3D Bathroom Designer that enables customers to create a customized bathroom layout, apply tiles, place showroom products, and inspect the design before purchasing.
3. To integrate the system with the showroom's inventory database to provide accurate real-time product availability and improve inventory management efficiency.
4. To provide an analytical dashboard that helps showroom management monitor sales trends, identify fast-selling and slow-moving products, and support data-driven business decisions.

2 Methodology

2.1 Agile Methodology – Scrum Framework

The development of the proposed system will follow the Agile Software Development Methodology using the Scrum framework. Scrum is selected because it supports iterative development, continuous improvement, flexibility, and effective team collaboration. Since the proposed system contains several interactive and advanced functionalities such as customer-based 3D bathroom designing, real-time inventory integration, and analytical dashboards, Scrum allows the development team to gradually implement and improve features through multiple development cycles called sprints.

Each sprint focuses on a set of prioritized functionalities, enabling the team to deliver a working product increment while continuously gathering feedback and making improvements throughout the software development lifecycle.

Before starting the sprint development process, the system requirements and functionalities are organized as Product Backlog Items. These backlog items include the major features that need to be developed. During each sprint planning session, selected Product Backlog Items are moved into the Sprint Backlog based on priority, complexity, and system dependencies. Each sprint is planned with a fixed duration of three weeks to ensure consistent progress, proper task allocation, and effective delivery of system functionalities.

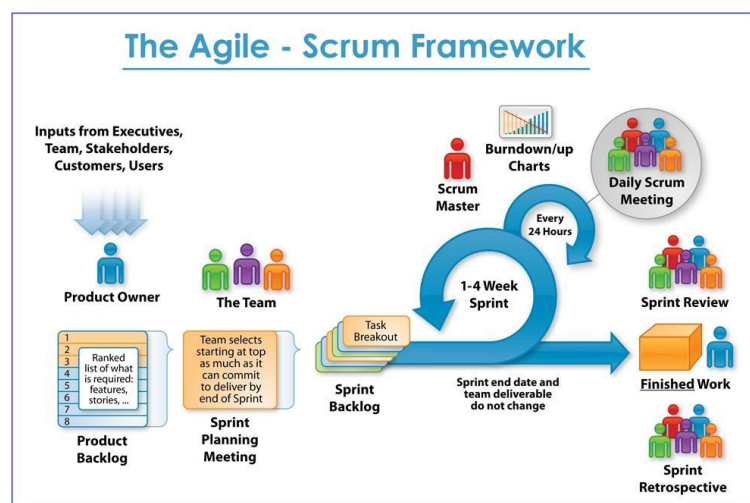


Figure 1 - The Agile–Scrum Framework

2.1.1 Sprint 1:

System Foundation, User Management, Product Browsing and Shopping Cart

1. Project Initiation and Release Planning:
 - Define the overall system scope and core functionalities.
 - Identify primary system users such as Guest Users, Registered Customers, and Administrators.
 - Finalize system architecture, database structure, and technology stack.
 - Define the basic product browsing and shopping cart workflow
2. Sprint Planning:
 - Plan user registration and authentication functionalities.
 - Plan product browsing, searching, and filtering mechanisms.
 - Plan shopping cart functionality and product selection workflow.
 - Plan initial frontend and backend setup.
3. Sprint Execution:
 - Develop frontend and backend architecture.
 - Implement customer registration, login, and logout functionalities.
 - Implement administrator authentication and secure admin access.
 - Develop product browsing, searching, and filtering functionalities.
 - Display product details including images, prices, discounts, brands, and available quantities.
 - Implement shopping cart management.
 - Enable users to add products to the cart, update quantities, and remove products.
4. Sprint Review:
 - Demonstrate user authentication and role-based access.
 - Demonstrate product browsing, filtering, and product detail viewing.
 - Demonstrate shopping cart management functionalities.
 - Gather feedback regarding usability, navigation, and customer interaction flow.

Sprint Backlog Items:

- User registration and authentication
- Administrator login and secure dashboard access
- Product browsing
- Product searching and filtering
- Product detail viewing
- Shopping cart management

2.1.2 Sprint 2:

Customer-Based 3D Bathroom Designer

1. Project Initiation and Release Planning:

- Finalize the scope of the Customer-Based 3D Bathroom Designer.
- Identify required 3D assets, bathroom layout options, door and window models, tile textures, and bathware item models.
- Define the customization features required to create a bathroom layout similar to the customer's actual space.

2. Sprint Planning:

- Plan bathroom shape, size, and dimension customization functionalities.
- Plan door and window selection and positioning features.
- Plan wall color, floor tile, and wall tile customization.
- Plan product placement, movement, rotation, replacement, and removal mechanisms.
- Plan measurement tool, 3D navigation, price summary, and design saving functionalities.

3. Sprint Execution:

- Develop the customized 3D bathroom design environment.
- Enable users to select bathroom shape and size and enter dimensions using feet or centimeters.
- Enable users to add doors and windows and adjust their positions.
- Implement wall color customization, floor tile selection, and wall tile placement with tile coverage height.
- Enable users to add bathware items and accessories such as bathtubs, wash basins, toilets, showers, mirrors, and other showroom products.
- Enable users to move, rotate, replace, and remove selected items inside the 3D bathroom.
- Implement zoom, rotation, and different angle inspection features.
- Implement ruler or measurement tool support for checking space and item placement distances.
- Automatically calculate and display the total price summary for selected tiles, bathware items, and accessories.

- Enable users to save customized 3D bathroom designs for future reference or purchasing decisions.

4. Sprint Review:

- Demonstrate bathroom layout creation and dimension customization.
- Demonstrate door, window, tile, wall color, bathware, and accessory customization.
- Demonstrate measurement, 3D navigation, price summary, and design saving features.
- Gather feedback on usability, customization accuracy, and visualization performance.

Sprint Backlog Items:

- Bathroom shape and dimension customization
- Door and window placement
- Wall color customization
- Floor tile and wall tile customization
- Bathware and accessory placement
- Product movement, rotation, replacement, and removal
- Measurement/ruler tool
- 3D zoom and rotation controls
- Total price summary
- Save customized design

2.1.3 Sprint 3:

Inventory Integration and Analytics Dashboard

1. Project Initiation and Release Planning:

- Finalize requirements for POS integration and analytical dashboard functionalities.
- Identify required sales, product, inventory, and stock data from the showroom system.
- Define key analytics outputs required for showroom management.

2. Sprint Planning:

- Plan real-time inventory synchronization with the POS database.
- Plan inventory monitoring and stock update functionalities.
- Plan analytics and decision-support features for administrators.
- Plan dashboard visualizations such as charts, graphs, and product comparisons.

3. Sprint Execution:

- Integrate the system with the showroom POS database.
- Implement real-time product availability and quantity updates.
- Develop inventory monitoring functionalities for administrators.
- Develop the analytics dashboard for sales and inventory analysis.
- Analyze fast-selling and slow-moving products.
- Generate analytical comparisons, charts, and graphs.
- Implement restocking suggestions and discount opportunity identification based on sales and inventory insights.

4. Sprint Review:

- Demonstrate POS database integration and real-time inventory updates.
- Demonstrate inventory monitoring and dashboard analytics.
- Review product performance comparisons and decision-support insights.
- Gather management feedback for further improvements.

Sprint Backlog Items:

- POS database integration
- Real-time inventory updates
- Inventory monitoring
- Sales trend analysis
- Product performance comparison
- Fast-selling and slow-moving item identification
- Restocking suggestions
- Discount opportunity identification
- Dashboard charts and graphs

2.1.4 Sprint 4:

Order Management, Package Visualization, Administrative Features, Notifications, Security and Final System Optimization

1. Project Initiation and Release Planning:

- Finalize remaining customer-side, administrator-side, notification, and security functionalities.
- Define order placement and order management workflow.
- Define package management and package visualization requirements.
- Prepare the system for final testing and deployment.

2. Sprint Planning:

- Plan order placement, order summary, and order confirmation functionalities.
- Plan package display and package visualization features.
- Plan administrative product, package, and discount management functionalities.
- Plan notification mechanisms and security enhancements.
- Plan final testing, bug fixing, and optimization activities.

3. Sprint Execution:

- Implement order placement and order confirmation functionalities.
- Generate order summaries and store order details securely.
- Develop predefined package display and package detail views.
- Enable package visualization within the Customer-Based 3D Bathroom Designer.
- Implement product management functionalities for administrators.
- Enable administrators to manage product categories, packages, and discounts.
- Implement customer order monitoring and inventory availability monitoring.
- Implement order confirmation and low-stock notification mechanisms.
- Apply security mechanisms to protect customer, product, order, and administrative data.
- Conduct system testing, bug fixing, performance optimization, and final improvements.

4. Sprint Review:

- Demonstrate order placement, order summary, and order management functionalities.
- Demonstrate package visualization and administrative management features.
- Review notification, security, and overall system performance.
- Finalize the deployable version of the system.

Sprint Backlog Items:

- Order placement
- Order summary and confirmation
- Secure order storage
- Package display and detail viewing
- Package visualization
- Product management
- Category, package, and discount management
- Customer order monitoring
- Low-stock and order notifications
- Security enhancements
- Final testing and optimization

By following the Agile Scrum methodology, the development team can incrementally develop and improve the system while adapting to changing requirements and feedback. This approach ensures continuous delivery of functional components, effective collaboration among team members, and successful implementation of both customer-oriented and business-oriented functionalities throughout the software development lifecycle. [7]

3 Requirements Identification

3.1 Functional and Non-functional requirements

3.1.1 Functional requirements

The functional requirements of the proposed system are divided into two main categories: Novelty Functional Requirements and General Functional Requirements. The novelty functional requirements represent the key innovative features of the system, while the general functional requirements support the core operations of the platform.

A. Novelty Functional Requirements

1. Customer-Based 3D Bathroom Designer

- Users should be able to create a customized 3D bathroom according to their own bathroom layout and preferences.
- Users should be able to select the bathroom shape and size and enter dimensions using feet or centimeters.
- Users should be able to select door and window designs and adjust their positions according to their actual bathroom layout.
- Users should be able to customize wall colors, floor tile designs, and wall tile placement by selecting specific walls and tile coverage height according to their preferences.
- Users should be able to add bathware items and accessories such as bathtubs, wash basins, toilets, showers, mirrors, and other available items from the showroom collection.
- Users should be able to move, rotate, replace, and remove selected items inside the 3D bathroom.
- Users should be able to use a ruler or measurement tool to measure spaces and item placement distances using the cursor while arranging items.
- Users should be able to zoom in, zoom out, and rotate the 3D bathroom view to inspect the design from different angles.
- The system should automatically calculate and display a total price summary for all selected tiles, bathware items, and accessories.
- Users should be able to save the customized 3D bathroom design for future reference or purchasing decisions.

2. Package Visualization Functionality

- Users should be able to view predefined product packages available in the showroom.
- Users should be able to view package details including included items, prices, and discounts.
- Users who physically visit the tile and bathware showroom should be able to compare the actual bathroom setup with the same package visualization available on the website, creating a consistent showroom experience.
- Users should be able to compare different packages based on included items, total price, discount details, and overall appearance.
- Users should be able to use package visualization to better understand the complete bathroom appearance before making a purchasing decision.

B. General Functional Requirements

1. User Registration and Authentication Functionality

- Guest users should be able to browse the system without creating an account.
- Customers should be able to register for an account before placing orders.
- Registered customers should be able to securely log in to their account.
- Registered customers should be able to securely log out from the system.
- Administrators should be able to securely access the admin dashboard

2. Product Browsing and Filtering Functionality

- Users should be able to browse product categories such as tiles, bathware, accessories, and packages.
- Users should be able to search products by name or category.
- Users should be able to filter products based on type, color, material, brand, size, and price range.
- Users should be able to view detailed product information including images, prices, discounts, and available quantity.

3. Shopping Cart and Order Management Functionality
 - Users should be able to add products to the shopping cart.
 - Users should be able to update product quantities or remove products from the cart.
 - Registered customers should be able to place orders for selected products.
 - The system should generate order summaries and confirmations.
 - The system should securely store customer order details.
4. Real-Time Inventory Management with POS Integration
 - The system should connect with the showroom's POS system database.
 - The system should display real-time product availability.
 - Product quantities should automatically update after purchases.
 - Administrators should be able to monitor inventory levels.
5. Analytics and Decision-Support Dashboard for Administrators
 - Administrators should be able to view fast-selling and slow-moving products.
 - The system should analyze sales trends and brand performance.
 - The system should provide recommendations for product restocking.
 - The system should suggest discount opportunities for low-demand products.
 - The system should display analytical insights using charts and graphs.
6. Administrative Management Functionality
 - Administrators should be able to add new products to the system.
 - Administrators should be able to update or remove existing products.
 - Administrators should be able to manage product packages and discounts.
 - Administrators should be able to monitor customer orders and inventory availability.
7. Notification Functionality
 - Customers should receive order confirmations and order updates.
 - Administrators should receive low-stock notifications.
 - Administrators should receive important order-related alerts related to orders and inventory status.

3.1.2 Non-functional requirements

1. Usability

- The user interface should be simple, interactive, and easy to understand for customers and administrators.
- The platform should be responsive and function properly on desktops, tablets, and mobile devices.
- Product browsing, package visualization, and Customer-Based 3D Bathroom Designer features should be easy to navigate with clear controls and labels.
- The system should provide user guidance such as alerts, notifications, and validation messages to help users complete tasks efficiently.
- The 3D bathroom designing features should allow users to customize layouts, compare products, and interact with the system without technical knowledge.

2. Security

- The system should use HTTPS to ensure secure communication between users and the server.
- Customer and administrator passwords should be securely stored using encryption or hashing techniques.
- The platform should implement role-based access control to separate customer and administrator functionalities.
- Administrative functionalities should only be accessible to authenticated admin users.
- The system should securely manage customer, product, order, and inventory-related information.
- The system should validate user inputs and protect against security threats such as SQL injection.

3. Compatibility

- The system should function properly on major web browsers including Chrome, Firefox, Safari, and Edge.
- The platform should support different screen sizes and resolutions through responsive web design.
- The Customer-Based 3D Bathroom Designer should work smoothly across supported devices and browsers.
- The system should integrate successfully with the showroom's POS system database for real-time inventory updates.

4. Maintainability

- The system should follow a modular architecture to simplify future development and maintenance.
- The codebase should follow clean coding standards and proper documentation practices.
- The frontend, backend, database, and 3D visualization components should be separated properly for easier maintenance.
- The system should allow easy updates to components such as product management, Customer-Based 3D Bathroom Designer, package visualization, and analytics dashboard.
- The system should support scalability for adding future features and product categories.

3.2 System requirements

3.2.1 Hardware Requirements

Development Environment:

- laptops with minimum 8GB RAM and quad-core processors
- Recommended: at least 16GB RAM for 3D visualization and AI model testing
- Internet connection for cloud database, GitHub, and deployment testing

Production Environment:

- Web server: 4GB RAM, 2-core processor, 100GB storage
- Database server: cloud-hosted MySQL/PostgreSQL database
- AI processing server: minimum 8GB RAM, GPU support recommended for SAM-based image processing
- Backup: automated daily database backups through cloud provider

3.2.2 Software Requirements

Development Tools:

- IDE: Visual Studio Code
- Version Control: GitHub
- Project Management: ClickUp
- UI/UX Design: Figma
- API Testing: Postman
- 3D Model Editing: Blender or Unity

Client Environment:

- Web browsers: Chrome, Firefox, Edge, Safari
- Mobile and desktop browser support
- Stable internet connection for loading 3D room designer and product data

3.2.3 Technologies

Front-end:

- Next.js
- TypeScript
- Tailwind CSS
- React Three Fiber
- Three.js
- Axios

Back-end:

- Node.js
- NestJS / Express.js
- RESTful API architecture
- JWT authentication
- Role-based access control

Database:

- MySQL
- POS database integration for real-time stock updates

3.3 User roles

1. Guest User

- Browse and search products available in the showroom
- View product details, prices, discounts, and available quantities
- Filter products based on category, type, color, brand, material, size, and price
- Access the Customer-Based 3D Bathroom Designer
- Create a customized 3D bathroom layout using shape, size, dimensions, doors, and windows
- Customize wall colors, floor tiles, and wall tile placement
- Add bathware items and accessories into the 3D bathroom
- View predefined product packages and package visualizations
- Add products to the shopping cart
- Register an account to place orders

2. Registered Customer

- Perform all functionalities available to guest users
- Create and manage their customer account
- Securely log in and log out from the system
- Place orders for selected products
- View order summaries and order history
- Receive order confirmations and notifications
- Manage shopping cart activities
- Save customized 3D bathroom designs for future reference or purchasing decisions

3. Administrator

- Securely access the admin dashboard
- Add, update, and remove products from the system
- Manage product categories, discounts, and packages
- Monitor product inventory and stock availability
- Access real-time inventory information integrated with the POS system
- View customer orders and manage order information
- Analyze sales trends and product performance
- Identify fast-selling and slow-moving products
- Receive restocking and discount recommendations through the analytics dashboard
- Manage the overall system and administrative functionalities

4 Project Plan

4.1 Gantt chart

Table 1 - Gantt Chart

Task	week														
	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Topic Identification & Requirements Gathering															
Requirements Analysis & specification															
Proposal Submission & Presentation															
Sprint 1															
Sprint 2															
Progress Presentation															
Sprint 3															
Sprint 4															
Final Report Submission and Presentation															
Documentation															

5 References

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